

Total No. of Questions : 8]

SEAT No. :

**PC1970**

[Total No. of Pages : 2

**[6353] 305**

**T.E. (Honors in Cyber Security)**

**ENTERPRISE ARCHITECTURE AND COMPONENTS**

**(2019 Pattern) (Semester - II) (310403)**

**Time : 2½ Hours]**

**[Max. Marks : 70**

**Instructions to the candidates:**

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicates full marks.
- 3) Draw neat diagram wherever necessary.
- 4) Assume suitable data, if necessary.

- Q1)** a) List the primary elements depicted in a component relationship diagram. [6]  
b) Explain the significance of the component model in software architecture. [5]  
c) Identify the key elements found in a component interaction diagram. [6]

OR

- Q2)** a) Describe how component relationship diagrams help in understanding system architecture. Provide an example to support your explanation. [9]  
b) Design a component interaction diagram for a real-time messaging application and explain your design choices. [8]

- Q3)** a) Define 'operational model' in the context of software engineering. [6]  
b) Explain the importance of understanding operational model design techniques in software development. [6]  
c) List three service qualities relevant to operational models. [6]

OR

- Q4)** a) Define the term 'operational pattern' and discuss its relationship with operational models. Provide examples to illustrate your explanation. [9]  
b) Discuss the advantages and disadvantages of various operational model design techniques in different project contexts. Use specific scenarios to illustrate your points. [9]

**P.T.O.**

- Q5)** a) Define “metadata” and “master data” in the context of data management. Explain the importance of clear terminology in data-related projects. [6]
- b) Identify and discuss service qualities crucial for effective metadata management. Explain how these qualities contribute to the success of metadata management initiatives. [6]
- c) Provide definitions for ‘business scenario’ and ‘component deep dive’ in the context of software engineering. Explain how these terms are relevant to system design and development. [5]

OR

- Q6)** a) Describe a business scenario where effective metadata management is critical. Explain how proper metadata management practices can benefit the scenario. [9]
- b) Describe three service qualities crucial for master data management systems. Explain how these qualities contribute to the reliability and consistency of master data across the organization. [8]
- Q7)** a) Explain the key principles and objectives of SABSA, COBIT, and TOGAF frameworks. Discuss the relationships between these frameworks and how they complement each other in enterprise architecture. [9]
- b) Explain how CMMI can be used to monitor, measure and report the progress of architecture development activities within an organization. Discuss the key process areas and metrics relevant to this task. [9]

OR

- Q8)** a) Provide a real-life example of an organization that has successfully implemented SABSA, COBIT, and TOGAF frameworks in its enterprise architecture. Discuss the benefits achieved and challenges encountered during the implementation process. [9]
- b) Outline the steps involved in using SABSA, COBIT, and TOGAF to develop an enterprise security architecture. Discuss the specific contributions of each framework in the process. [9]

